
Adaptive Inference Scheduling for Diffusion-Based MRI Reconstruction

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Abstract

Accelerated MRI reduces scan time by acquiring only a fraction of k-space measurements, making high-quality reconstruction from incomplete data a key challenge. Diffusion models achieve strong reconstruction performance but use a fixed inference budget for all samples, wasting computation on slices that have already converged. We propose adaptive inference scheduling, a per-sample early stopping method for DDIM reconstruction that monitors the intermediate clean-image estimate and halts inference once it stabilises. The method operates entirely at inference time and requires no retraining. Experiments on the Calgary-Campinas-359 dataset at $R = 4$ acceleration show that adaptive inference preserves reconstruction quality close to DDIM-100 while reducing denoising steps by 25% and wall-clock time by 22%, with individual slices stopping at different steps, consistent with per-sample adaptation.

1 Introduction

Magnetic resonance imaging (MRI) is widely used for medical diagnosis but requires long acquisition times, typically 20 to 60 minutes, increasing cost, reducing patient throughput, and increasing susceptibility to motion artifacts GharehMohammadi and Sebro [2024]. Accelerated MRI addresses this by acquiring only a subset of k-space measurements, but undersampling introduces aliasing artifacts, making accurate reconstruction essential for clinical utility.

Diffusion models have shown strong reconstruction performance by iteratively denoising from Gaussian noise conditioned on undersampled measurements Webber and Reader [2024]. However, existing approaches apply a fixed inference step budget uniformly across all samples regardless of complexity. In practice, MRI slices vary substantially in anatomical structure: simpler slices may converge substantially earlier than slices containing fine structural detail. Applying the same number of denoising steps to every slice therefore wastes computation on samples that have already stabilised.

We investigate whether adaptive per-sample stopping can reduce this computational cost without sacrificing reconstruction quality. Our approach monitors the model’s intermediate clean-image estimate at each DDIM step and halts inference once that estimate stabilises, returning it directly as the final reconstruction. Stopping is performed independently per sample and requires no retraining or architectural modification. On the Calgary-Campinas-359 dataset Souza et al. [2018] at $R = 4$ acceleration, adaptive inference preserves reconstruction quality close to DDIM-100 while reducing denoising steps by 25% and wall-clock time by 22%, with different slices stopping at different steps, consistent with per-sample adaptation.

2 Related Work

Deep learning for accelerated MRI reconstruction. Early methods used unrolled optimisation networks interleaving learned priors with data-consistency updates; MoDL Aggarwal et al. [2019] is a representative example. Later convolutional networks achieved strong performance but often produced overly smooth reconstructions. Generative approaches including diffusion models were subsequently introduced to recover sharper anatomical detail Chung and Ye [2022].

Diffusion models for MRI reconstruction. Ho et al. [2020] introduced DDPMs for iterative image generation. Diffusion models have since been applied to image inpainting Lugmayr et al. [2022], general image restoration Bansal et al. [2023], and MRI reconstruction with k-space data consistency Chung and Ye [2022]. Other work incorporated edge-aware priors to preserve anatomical boundaries Majeed et al. [2025]. Our work instead focuses on reducing the inference cost of diffusion MRI reconstruction.

Efficient diffusion inference. Standard DDPM sampling may require hundreds or thousands of denoising steps, making inference computationally expensive. DDIM Song et al. [2021] reduces this cost using deterministic sampling with substantially fewer steps. Additional acceleration approaches include distillation, consistency models, and flow matching Lipman et al. [2023]. Existing methods reduce computation uniformly across samples, whereas our approach adaptively selects the stopping point for each individual MRI slice.

Adaptive stopping methods. Adaptive computation has been explored in early-exit networks and iterative solvers, but not, to our knowledge, in diffusion-based medical image reconstruction. Unlike learned gating methods, our approach uses only the model’s own intermediate estimate $\hat{x}_0^{(t)}$ together with a simple threshold-based stopping rule, requiring no additional training or architectural modification.

3 Method

3.1 Problem Formulation

Let $\mathbf{y} \in \mathbb{C}^N$ denote the undersampled k-space measurements where $\mathbf{y} = \mathbf{M}\mathcal{F}(\mathbf{x})$, with $\mathbf{x} \in \mathbb{R}^{H \times W}$ the unknown image, $\mathcal{F}$ the 2D Fourier transform, and $\mathbf{M} \in \{0, 1\}^N$ a binary undersampling mask. At $R = 4$, approximately 25% of k-space measurements are retained.

3.2 Diffusion Model and DDIM Background

Ho et al. [2020] introduced DDPMs, which learn to reverse a gradual Gaussian noising process:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (1)$$

via a U-Net $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \mathbf{c}, t)$ conditioned on the undersampled image $\mathbf{c}$. At inference we use deterministic DDIM Song et al. [2021] with $\eta = 0$, which produces an intermediate clean-image estimate at each step:

$$\hat{\mathbf{x}}_0^{(t)} = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \mathbf{c}, t)}{\sqrt{\bar{\alpha}_t}}, \quad (2)$$

and updates the latent via:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{\mathbf{x}}_0^{(t)} + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \mathbf{c}, t). \quad (3)$$

Because $\eta = 0$ adds no stochastic noise, the trajectory of $\hat{\mathbf{x}}_0^{(t)}$ tends to stabilise as sampling progresses, enabling per-sample convergence monitoring.

3.3 Adaptive Stopping Criterion

We define the per-sample relative change at step s as:

$$\delta_s = \frac{\|\hat{\mathbf{x}}_0^{(s)} - \hat{\mathbf{x}}_0^{(s-1)}\|_2}{\|\hat{\mathbf{x}}_0^{(s-1)}\|_2 + \epsilon}, \quad (4)$$

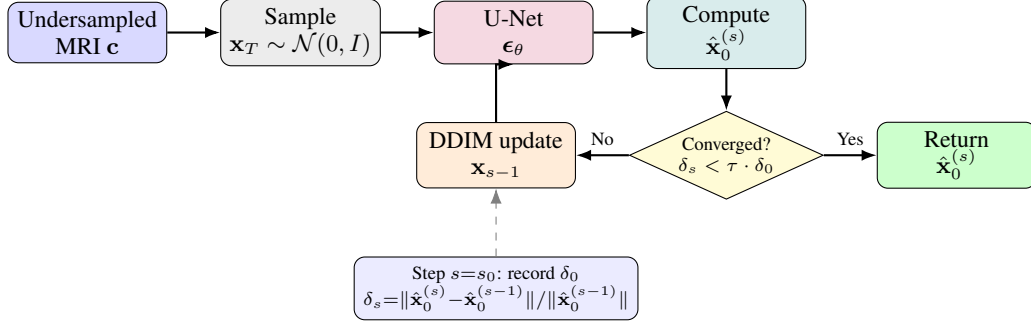


Figure 1: Adaptive inference pipeline. The U-Net iteratively denoises conditioned on the undersampled MRI, computing relative change δ_s in $\hat{\mathbf{x}}_0^{(s)}$. Once $\tilde{\delta}_s$ falls below $\tau \cdot \delta_0$ for $P = 3$ consecutive steps, inference halts and the current clean-image estimate is returned directly.

where ϵ is a small constant for numerical stability.

Baseline recording. We run the first $s_0 = 30$ steps without checking convergence. At step s_0 we record $\delta_0 = \delta_{s_0}$ as the per-sample baseline.

Relative threshold. From step $s_0 + 1$ onward, we compute a smoothed signal $\tilde{\delta}_s = \frac{1}{W} \sum_{j=0}^{W-1} \delta_{s-j}$ with $W = 3$. A sample is considered converged when:

$$\tilde{\delta}_s < \tau \cdot \delta_0, \quad (5)$$

where $\tau = 0.35$, indicating the reconstruction has stabilised relative to its early denoising behaviour. The threshold was selected empirically from validation convergence behaviour; supporting diagnostics are in section A and validated via ablation in section 4.6.

Patience and early return. To avoid stopping on transient dips, we require eq. (5) to hold for $P = 3$ consecutive steps. We return $\hat{\mathbf{x}}_0^{(s)}$ directly rather than $\mathbf{x}_s$, as the latter remains a noisy latent variable.

Per-sample independence. Stopping decisions are made independently per sample, allowing simpler slices to terminate earlier while harder slices continue denoising. The method operates entirely at inference time and requires no retraining.

Algorithm 1 Adaptive DDIM Sampling

```

1: Initialize  $\mathbf{x}_T \sim \mathcal{N}(0, I)$ ,  $\hat{\mathbf{x}}_0^{(1)} \leftarrow \mathbf{0}$ ,  $\delta_0 \leftarrow \text{None}$ , patience  $\leftarrow 0$ 
2: for  $s = 2$  to  $S_{\max}$  do
3:   Predict  $\hat{\mathbf{x}}_0^{(s)}$ ; compute  $\delta_s$  via eq. (4)
4:   if  $s = s_0$  then  $\delta_0 \leftarrow \delta_s$ 
5:   end if
6:   if  $s > s_0$  then
7:      $\tilde{\delta}_s \leftarrow \frac{1}{W} \sum_{j=0}^{W-1} \delta_{s-j}$ 
8:     if  $\tilde{\delta}_s < \tau \delta_0$  then
9:       patience  $\leftarrow$  patience + 1
10:    else
11:      patience  $\leftarrow 0$ 
12:    end if
13:    if patience  $\geq P$  then return  $\hat{\mathbf{x}}_0^{(s)}$ 
14:    end if
15:  end if
16:  Apply DDIM update (eq. (3))
17: end for
18: return  $\hat{\mathbf{x}}_0^{(S_{\max})}$ 

```

Table 1: Reconstruction quality and efficiency on 700 validation slices ($R=4$, CC-359). Mean $\pm$ std reported.

Method	SSIM $\uparrow$	PSNR $\uparrow$	NMSE $\downarrow$	Steps	Time (s)
DDIM-25	0.208 ± 0.081	13.62 ± 1.51	0.287 ± 0.146	25.0	1.054
DDIM-50	0.322 ± 0.137	14.97 ± 1.91	0.219 ± 0.134	50.0	2.106
DDIM-100	0.366 ± 0.129	15.15 ± 1.96	0.213 ± 0.128	100.0	4.241
Adaptive	0.350 ± 0.117	15.32 ± 2.07	0.207 ± 0.125	75.0	3.300

3.4 Model Architecture

The denoising network is a U-Net conditioned on the undersampled image and timestep embedding, following Ho et al. [2020]. The encoder compresses 256×256 inputs to 32×32 across four levels with channel multipliers $[1, 2, 4, 8]$. Self-attention is applied at the two deepest levels; skip connections restore spatial detail in the decoder. The network has 48.4 million parameters.

4 Experiments

4.1 Dataset and Experimental Setup

Dataset. We use single-channel magnitude reconstructions derived from the Calgary-Campinas-359 (CC-359) dataset Souza et al. [2018]. Each subject provides 170 axial slices at 256×256 ; discarding the first and last 50 edge slices leaves 70 central slices per subject, giving 1,610 training and 700 validation slices. We simulate $R = 4$ acceleration using a random undersampling mask with 8% centre fraction, randomising the mask seed per epoch to prevent memorisation.

Implementation. We train with AdamW Loshchilov and Hutter [2019] at learning rate 10^{-4} with cosine annealing to 10^{-6} over 200 epochs, batch size 16, and gradient clipping at 1.0. The best checkpoint (epoch 79, validation SSIM = 0.311) is selected by DDIM-50 inference every 10 epochs.

4.2 Baselines

We compare against three fixed-step DDIM baselines using the same model weights: **DDIM-25** (25 steps), **DDIM-50** (50 steps), and **DDIM-100** (100 steps, primary baseline). All use $\eta = 0$. The adaptive method uses a maximum of 100 steps, making the comparison with DDIM-100 fair.

4.3 Evaluation Metrics

We report **SSIM** $\uparrow$, **PSNR** (dB) $\uparrow$, and **NMSE** $\downarrow$ on 700 validation slices, plus mean steps and wall-clock time for the adaptive method.

4.4 Main Results

Adaptive inference achieves PSNR of 15.32 dB and NMSE of 0.207, both better than DDIM-100, while using 75.0 steps on average – a **25.0%** step reduction and **22.2%** wall-clock reduction. SSIM of 0.350 is slightly below DDIM-100 (0.366), a 4.4% decrease. Adaptive therefore improves the efficiency–quality tradeoff, substantially outperforming DDIM-50 while remaining cheaper than DDIM-100. Crucially, 690 of 700 slices (98.6%) stopped before the full budget, confirming per-sample adaptation.

Figure 1 illustrates the adaptive inference pipeline. Figure 3 shows qualitative reconstructions: adaptive produces visually comparable brain structure to DDIM-100 while DDIM-25 shows noticeably degraded contrast and detail.

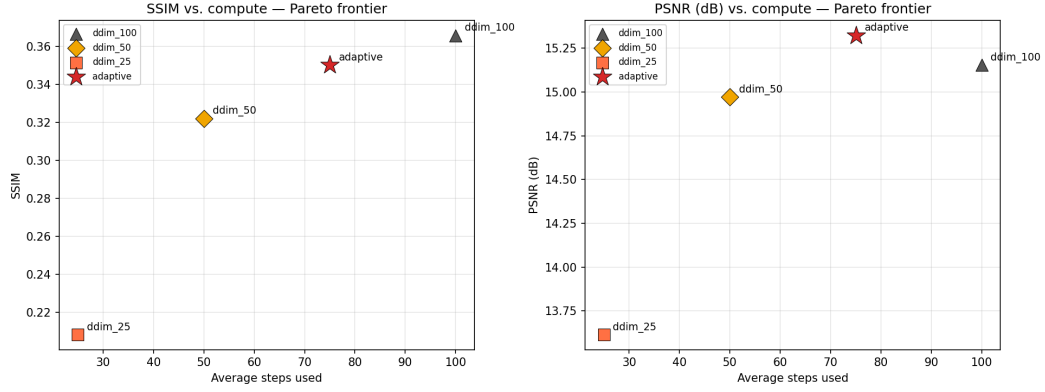


Figure 2: Quality–compute Pareto frontier. Adaptive (red star) substantially improves over DDIM-50 while remaining cheaper than DDIM-100.

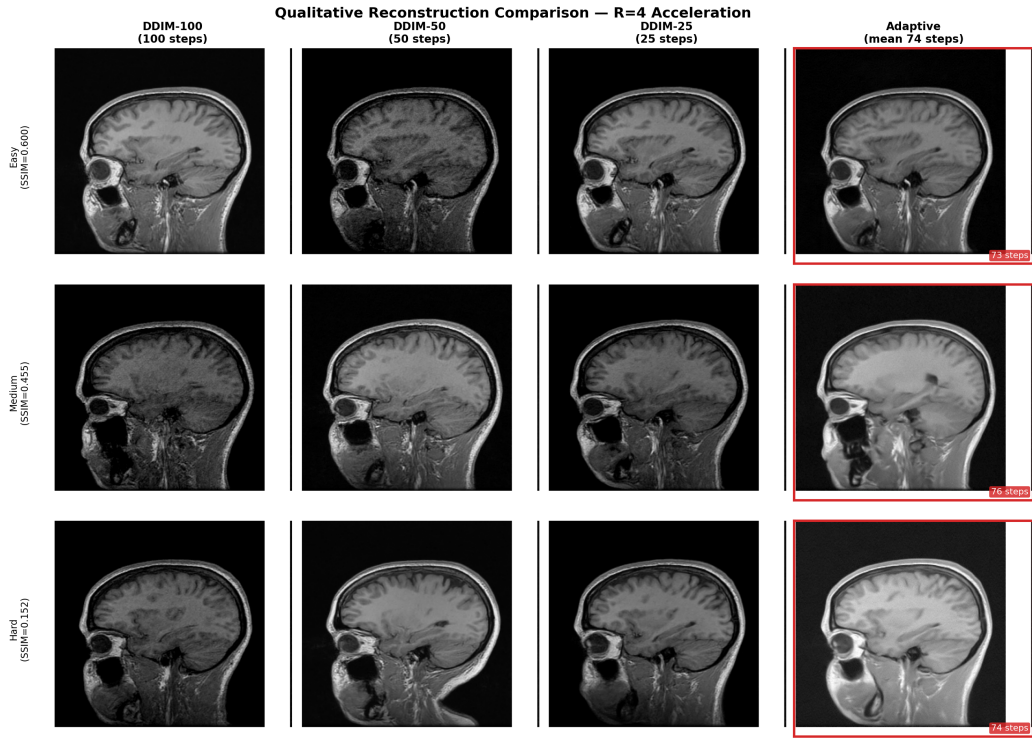


Figure 3: Qualitative comparison on three validation slices ($R = 4$). Adaptive uses different stopping steps per slice (73–76) while producing reconstructions visually comparable to DDIM-100.

4.5 Per-Sample Analysis

Stopping steps range from 63 to 100 (mean 75.0, median 74.0, std 4.0), indicating per-sample variation in convergence behaviour. Per-sample SSIM shows a tight band around the adaptive mean (std 0.117) regardless of stopping step, indicating consistent reconstruction quality. Full diagnostics are in section A.

4.6 Ablation Study

At $\tau = 0.35$ we obtain a 25% reduction in denoising steps while maintaining reconstruction quality close to DDIM-100. Larger thresholds reduce compute further but increase instability and quality

Table 2: Ablation over threshold τ on 200 validation slices.

Threshold τ	SSIM $\uparrow$	PSNR $\uparrow$	NMSE $\downarrow$	Steps
0.20	0.361	15.04	0.217	100.0
0.25	0.369	15.21	0.206	99.9
0.35 (ours)	0.367	15.59	0.193	74.8
0.40	0.370	15.95	0.173	67.4
0.50	0.326	15.47	0.200	55.5
DDIM-100	0.366	15.25	0.205	100.0

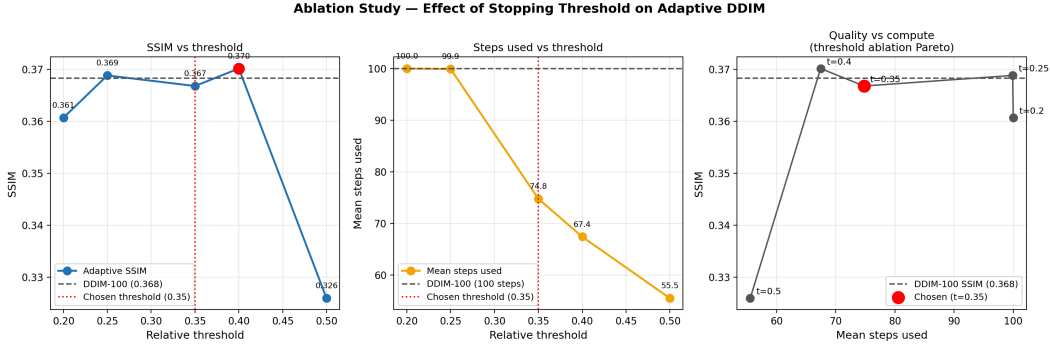


Figure 4: Ablation over τ . *Left*: SSIM stable from $\tau = 0.20$ to 0.40 , collapses at 0.50 . *Centre*: Steps decrease monotonically. *Right*: $\tau = 0.35$ on the efficient frontier at 25% fewer steps.

degradation. For $\tau \leq 0.25$ the criterion rarely triggers, whereas at $\tau = 0.50$ quality drops noticeably (55 steps). Although $\tau = 0.40$ achieves slightly stronger mean metrics on this 200-slice subset, it also produces more aggressive early stopping. The adaptive SSIM values on this smaller ablation subset are slightly higher than those observed on the full 700-slice evaluation, indicating moderate subset-dependent variance in reconstruction difficulty. We therefore select $\tau = 0.35$ as a more conservative operating point that better preserves reconstruction quality across the full validation set.

5 Discussion

Diffusion MRI reconstruction exhibits diminishing returns at late denoising stages: many slices stabilise substantially before the full budget is exhausted. Adaptive stopping exploits this to reduce unnecessary computation while preserving reconstruction quality close to DDIM-100. Because the method operates entirely at inference time, it integrates into existing pipelines without retraining.

Absolute reconstruction quality (SSIM ≈ 0.35) remains below state-of-the-art methods on CC-359 Chung and Ye [2022], which use larger datasets and complex-valued k-space data consistency projection. This does not affect the validity of our contribution, which improves efficiency relative to the fixed-step baseline regardless of absolute quality level. Interestingly, the best reconstruction checkpoint occurred at epoch 79 even though the diffusion training loss continued decreasing afterwards, suggesting that diffusion training loss and downstream reconstruction quality may become decoupled during later training stages. The threshold τ was selected empirically and may require retuning under different sampling patterns or architectures.

Future work includes evaluation at $R = 8$, complex-valued reconstruction with data consistency projection, and learned stopping criteria. Applying adaptive stopping to flow matching models Lipman et al. [2023] is particularly promising, as straighter sampling trajectories are expected to yield smoother convergence signals and earlier reliable stopping.

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A Convergence Diagnostics and Per-Sample Analysis

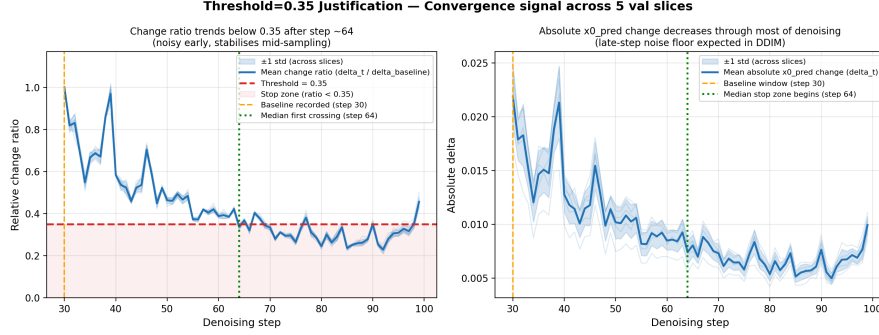


Figure 5: Convergence signal across 5 validation slices. *Left*: The mean change ratio δ_s/δ_0 (defined in eq. (4)) consistently falls below 0.35 between approximately denoising steps 64–75 (median first threshold crossing at step 64, crossing range 64–70; distinct from the full stopping range of 63–100 across all 700 validation slices), confirming that $\tau = 0.35$ is a natural operating point. The signal stabilises during the mid-to-late denoising stages; in practice inference halts at the first crossing and the post-crossing trajectory is shown for illustration only. *Right*: Absolute change δ_s decreases through most of denoising, with a late-step noise floor consistent with deterministic DDIM.

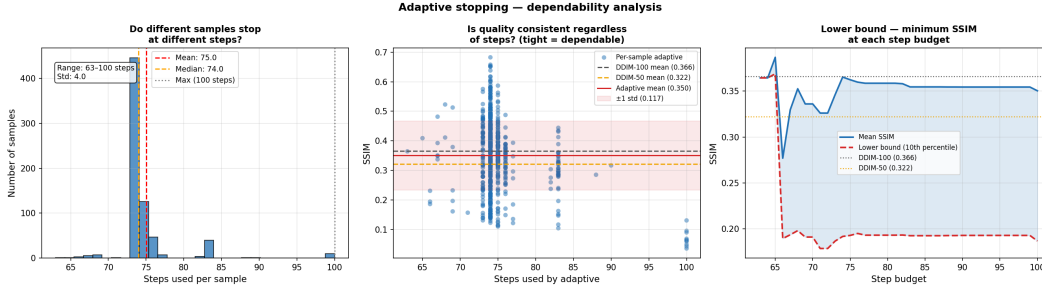


Figure 6: Per-sample adaptive stopping analysis on 700 validation slices. *Left*: Denoising step distribution (range 63–100, mean 75.0, median 74.0, std 4.0), consistent with per-sample adaptation. *Centre*: SSIM vs. denoising steps used, showing a consistent quality band regardless of stopping step, directly confirming dependable reconstruction quality. *Right*: Lower bound curve (10th percentile SSIM) – the lowest-quality tail remains relatively stable, though below the DDIM-50 mean.

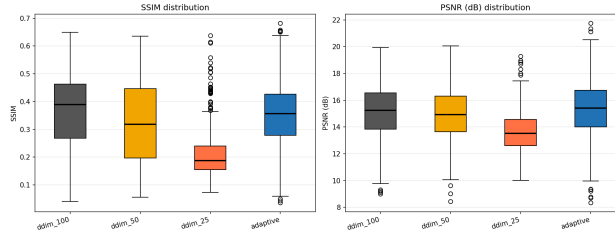


Figure 7: SSIM and PSNR distributions across 700 validation slices. Adaptive inference maintains a comparable distributional spread to DDIM-100 while using approximately 25% fewer denoising steps on average, confirming that quality consistency is preserved across the full validation set.

B Implementation Details

All experiments were implemented in PyTorch and run on a single NVIDIA A100 40GB GPU. The diffusion model uses a cosine noise schedule over $T = 1000$ timesteps, sampled using 100 uniformly

spaced DDIM timesteps [10] with $\eta = 0$. Training used the AdamW optimiser [7] with learning rate 10^{-4} , cosine annealing to 10^{-6} , batch size 16, and gradient clipping at 1.0 over 200 epochs. During evaluation, undersampling masks were generated using fixed random seeds to ensure reproducibility across methods. Wall-clock time was measured as the mean inference time per batch of 4 slices, averaged over 175 evaluation batches. All results are reported on the same fixed validation split of 700 slices from 10 subjects not seen during training.

Table 3 summarises the hyperparameters of the adaptive stopping criterion. These values were fixed for all experiments; only τ was varied in the ablation study (section 4.6).

Table 3: Adaptive stopping hyperparameters used in all experiments.

Parameter	Symbol	Value
Baseline window step	s_0	30
Smoothing window size	W	3
Patience	P	3
Relative threshold	τ	0.35
Maximum steps	$S_{\max}$	100

Code. The code for this project is available at: <https://github.com/up-the-ladder/adaptive-inference-diffusion-mri>